

Provisional Patent Application 16 - Liana Banyan Corporation

Substrate-Routed Memory Expansion, Multi-Layer Authority Recursion, Social-Authority DAG Governance, Cooperative Federation Library Architecture, and Cooperative AI Substrate Primitives

Jonathan Jones, Sole Inventor

May 1, 2026

Provisional Patent Application 16

Liana Banyan Corporation

Cross-References to Related Applications

Filing Manifest — Innovation Cluster Summary

Cooperative Defensive Patent Pledge (#2260) — Filing Posture

Path B Discipline (Proof Before Claim)

Filing Window and Priority

Field of the Invention

Background of the Invention

Summary of the Invention

Brief Description of Figures

Detailed Description of the Invention

Claims

Abstract

Filing Gate Status

Provisional Patent Application 16

Liana Banyan Corporation

Applicant: Liana Banyan Corporation **Entity Type:** Wyoming C-Corporation (Small Entity) **EIN:** 41-2797446 **Inventor:** Jonathan Jones (sole inventor) **Filing Type:** Provisional Patent Application (USPTO Patent Center) **Estimated Filing Fee:** \$65 (Small Entity, provisional) **Priority Date Goal:** 2026-05-01 (KN071 FIRE-TODAY, Founder pre-authorization BP006 turn 6) **Conversion Deadline:** May 1, 2027 (12-month provisional window) **Filing path:** USPTO Patent Center direct — Founder discretion.

Title of Invention: *Method and System for Substrate-Routed Memory Expansion, Multi-Layer Authority Recursion, Social-Authority DAG Pheromone-Anchored Decision Governance, and Cooperative Federation Library Architecture in AI Substrate Platforms*

Filed under: #2260 Cooperative Defensive Patent Pledge (Liana Banyan Corporation, B098).

PUBLICATION GATE HARD: This specification is **internal-only** until Founder USPTO submission. Knight scope: build and stage. Founder scope: review and authorize USPTO submission. Fire Control per #2330; Founder authorized KN071 per BP006 turn 6.

Cross-References to Related Applications

This application claims priority benefit from the following related provisional applications, all filed by the same applicant (Liana Banyan Corporation) with the same sole inventor (Jonathan Jones):

- **Provisional Application No. 64/036,646**, filed April 12, 2026 (Provisional 13) — *Cooperative-Platform AI Memory Substrate: Cathedral Effect Cross-Vendor Uniformity, Pheromone Substrate Stigmergic Indexing, and Related Primitives*
- **Provisional Application No. 64/052,602**, filed April 29, 2026 (Provisional 14) — *Cooperative-Platform AI Memory Infrastructure — Multi-Thousand-Innovation Formal Ledger, Substrate Discipline Primitives, and Cooperative Sovereign Substrate Architecture*
- **Provisional Application No. 64/052,618**, filed April 29, 2026 (Provisional 15) — *Cooperative-Platform AI Memory Infrastructure — Agent-Spawn Boundary Pre-Injection, Lossless Vendor-Layer Tablet Capture, Corpus-Alias Registry-Keyword-Extension, and Substrate Discipline Primitives*

The architecture disclosed in this application composes with, extends, and provides reduction-to-practice evidence for innovations initiated in those prior applications. This filing constitutes the catch-all consolidation of all architectural primitives ratified across BP002 through BP006 that are not yet covered by Applications 64/036,646, 64/052,602, or 64/052,618.

Filing Manifest — Innovation Cluster Summary

#	Innovation Cluster	Short Description	Priority
2336	Substrate-Routed Memory Expansion	Agent memory index scales O(relevant-topics-per-task) not O(total-topics); Wrasse-routed delivery of content only when topic surfaces; 65%	CROWN JEWEL

#	Innovation Cluster	Short Description	Priority
2337	Ring of Three Golden Eblets Authority Structure	MEMORY.md compaction empirical receipt (KN042) Top-of-substrate must/must-not authority three-component structure (Canon/Platform/Project); recursive instantiation across N authority layers; Marked Exception non- overrideable routing	CROWN JEWEL
2338	Multi-Layer Authority Recursion	Substrate-as-a-service recursive deployment; each successive authority layer instantiates own Ring of Three with own brand/stamps/QR anchored to own IP Ledger; rules- stack compounds	CROWN JEWEL
2339	Social-Authority DAG + Pheromone-Anchored Decisions	Cross-cutting social- authority overlays (Guild/Tribe/Family/Helm) on recursive substrate hierarchy; sovereignty over decision-mechanism (HOW) and decision- outcome (WHOM); decisions as Pheromone records	CROWN JEWEL
2340	AGPL Cathedral Free / Federation Library Cooperative Architecture	Dual-value-prop: base substrate AGPL v3 free + full-featured; cross- member cooperative library opt-in at \$5/year; network-effect compounds with member count; Pledge prevents weaponization	HIGH
2341	LB Frame Broadcast Funnel (5-Stage Conversion)	5-stage substrate adoption funnel: Install→Demo→User- data-Test→Join→Send; Cathedral Effect makes all five stages substrate- verifiable	HIGH
2342	Knight-Has-No-Shadows / Bishop-Only Persistent-Substrate Operations	Persistent-substrate AI agent class (with hook architecture / persistent daemons = Shadows) vs per-session agent class; LIGHTHOUSE 8/2	HIGH

#	Innovation Cluster	Short Description	Priority
2343	Augur Living Gate (Substrate-Event-Driven Freshness)	instrumentation requires Shadows; Foreman/Sergeant asymmetry Substrate-event-driven freshness gate replaces clock-TTL: checks Pheromone-substrate-write timestamps vs last-consult timestamp; gate fires only on actual state change; hard ceiling fallback	HIGH
2344	Catechist Scribe (SessionStart Discipline Grading)	AI-agent session-discipline auto-grading via SessionStart hook; reads first-N tool calls; emits per-rule PASS/WARN/FAIL grades with cryptographic signature (Chronos HMAC-SHA256)	HIGH
2345	The Shadow (Persistent Background Watcher Daemon)	Persistent background watcher daemon spawned at AI-agent SessionStart + killed at SessionEnd; cross-monitor capture at session boundaries; brand-class wrapper for ALL persistent background tasks	HIGH
2346	Augur PostToolUse Audit + Stone Tablet Supersede Pattern	Converting AI-agent discipline gates from PreToolUse-blocking to PostToolUse-audit + Stone Tablet supersede: write succeeds, audit flags violation, supersede-pointer written for Bishop reconciliation	MEDIUM
2347	Walkaround Universal Install-Time Preflight Primitive	Universal AI substrate install-time preflight that emits empirical receipt as the installation completion demonstration; self-verifying install replaces marketing claims with measurable green-check output	CROWN JEWEL
2287	Partnership-Stake Downstream-IP-	IP-reversion clause triggered on substrate-	HIGH

#	Innovation Cluster	Short Description	Priority
2288	Reversion Clause	dependency threshold crossing; anti-enshittification framing; Substrate-Dependency Classifier adjudication	MEDIUM
	Cephas Pre-Submission Voting + Acceptance Rewards	Cooperative content-amplification with Marks-stake voting; Six-Degrees activation; acceptance-reward funded from patent revenue connected to content subject matter	
2289	Substrate-Dependency Classifier	Transparent algorithmic adjudication of substrate-dependency level via tokenization-and-alias-match against substrate-dependency graph; Tier A/B/C classification with empirical gates	HIGH
2348	Stay-Warm Discipline / Round the Horn / Pre-Staging Architecture	K-prompt bundling at session-open; Wrasse pre-injection of canonical answers; Detective live-update; pre-staged operator interfaces; substrate freshness maintained across session boundaries	HIGH
2349	Detective→Wrasse Live-Update Path	Phase-0 pheromone-substrate fast-path for Detective query results feeding back into Wrasse registry; 74ms build / 1ms query at 1593 records / 49:1 retrieval ratio vs filesystem text-matching	HIGH
2350	BRIDLE v11 Cross-Agent Edict Propagation	Cross-agent cooperative-substrate edict propagation across heterogeneous agent classes (Bishop/Knight/Pawn); edict propagation into Wrasse vocabulary (WV-001 through WV-004)	MEDIUM

Total innovations in this filing: 18 innovations, 15+ empirical receipts across BP002–BP006.

Cooperative Defensive Patent Pledge (#2260) — Filing Posture

All innovations in this provisional are filed under the **LB Cooperative Defensive Patent Pledge** framework (originated B098, propagated through every provisional filing since Prov 13). The Pledge’s operating principle:

“We file the IP so no one can lock it away, and we pledge to never use it offensively against anyone building on the cooperative substrate.”

Specific Pledge commitments for Prov 16 scope:

Every nonprofit, cooperative, and academic institution receives use rights to every architecture in this filing **free, forever**, on nothing more than an IRS-verified EIN (or international equivalent). No membership required, no fee, no approval gate.

The Pledge is administered by Liana Banyan Corporation governance per the Structural Bylaws. This filing protects the architecture from third-party enclosure; it does not claim exclusivity over cooperative-sector use.

Claim framing discipline: Every independent claim in this filing is drafted to be **maximally broad** while remaining defensible on Path B empirical-receipt grounds. The cooperative-sector use exemption attaches at the Pledge layer; the claims themselves protect the full architecture including commercial implementations, ensuring the Pledge is the difference between “free for cooperatives” and “free for everyone,” not between “filed” and “unfiled.”

Path B Discipline (Proof Before Claim)

All claims in this filing are grounded in empirical receipts per the Path B discipline (B130 Founder-ratified, project_path_b_proof_before_claim_b130.md): **build the reduction-to-practice artifact first; file the provisional with the working build as Section 2 evidence.**

No claim in this filing is a “glorified pitch deck” (Pawn Q2 enabling-disclosure test): every independent claim cites either a committed code artifact, a test receipt with pass count, or a measured empirical outcome that constitutes reduction-to-practice evidence under 35 U.S.C. § 102.

Innovation	Empirical Anchor	Call Sign /
#2336 Substrate-Routed Memory Expansion	KN042 Pod O	
	BP005 — MEMORY.md 405→141 lines (65%), 8/8 triggers, 33/33 tests	commit 0696f31
#2337 Ring of Three Golden Eblets	BP005 base ratification — 3 Golden Eblets 282	project_ring_of_three_golden_eblets_de

Innovation	Empirical Anchor	Call Sign /
	lines + 450+ line federation canon	
#2338 Multi-Layer Authority Recursion	KN046 Furnace Multi-Tenancy + KN045 Layer Addressing + KN049 Cycle Prevention — 34/34 tests	Pod Q BP005
#2339 Social- Authority DAG	KN050 Pheromone- Anchored Decision Schema (Pod R BP005) — 27/27 tests; full lifecycle	Pod R BP005
#2340 AGPL / Federation Library	KN057 L5 simulation — 50/50 tests; 5-layer compound 35.72× = 97.2% savings	commit ec1bda0
#2341 LB Frame Broadcast Funnel	Pod T LibrarianMedallion 35/35 + Pod Y LibrarianPage 20/20 + Pod AA Bounty Posters 36/36	KN061/KN064/KN066
#2342 Knight- Has-No-Shadows	KN037 v2 Shadow daemon 13/13 + KN067 Phase-E trigger 19/19	ba2131f / dedd541
#2343 Augur Living Gate	KN038 Pod N BP004 — 18/18 tests + T12 empirical PROOF gate stays open beyond TTL	commit 98d1745
#2344 Catechist Scribe	KN036 Pod M BP004 — 41/41 tests (22 rule + 19 integration)	commit 7fe3ebd
#2345 The Shadow	KN037 v2 Pod M BP004 — 13/13 tests + live smoke test green	commit ba2131f
#2346 Augur PostToolUse Audit	KN043 Pod P BP005 — 8/8 tests; PreToolUse blocking ENDED	commit 7701c84

Innovation	Empirical Anchor	Call Sign /
#2347 Walkaround Preflight	BP006 turn 3: 26 PASS / 1 WARN tabular live output; BP006 turn 4-5 Founder ratification	Walkaround.ps1
#2349 Detective→Wrasse Live-Update	K550 — 74ms build / 1ms query at 1593 records / 49:1 ratio vs grep	B134 turn 14 receipt
#2350 BRIDLE v11 Edict Propagation	WV-001 through WV-004 propagated into Wrasse vocabulary + Pawn dispatch directives	B134 edict activation

Filing Window and Priority

This provisional application is filed **2026-05-01** (BP006 FIRE-TODAY override, Founder pre-authorization per “We will file Prov 16 today”). The 12-month conversion deadline for this filing is **May 1, 2027**. The non-provisional conversion path activates KN071 spec as the formal-spec-PDF base (deadline 2026-11-26 for the conversion planning window).

Chain context: 15 provisional applications were filed by Liana Banyan Corporation preceding this filing (Prov 1 through Prov 15); this filing is the 16th.

The sequential provisional chain protecting the cooperative-platform AI memory substrate architecture: - Prov 13: 64/036,646 (April 12, 2026) — Cathedral Effect, Pheromone Substrate baseline - Prov 14: 64/052,602 (April 29, 2026) — Multi-thousand-innovation formal ledger cluster - Prov 15: 64/052,618 (April 29, 2026) — B132-B133 primitive cluster (Wrasse, Stone Tablet, Fire Control, Call Sign, Cut In Half / Both-Feet, Multi-Cylinder Dispatch) - **Prov 16: THIS FILING (May 1, 2026)** — BP002-BP006 architectural ratifications catch-all

Field of the Invention

This invention relates to cooperative multi-agent artificial intelligence substrate architectures; specifically, to methods and systems for memory expansion via substrate-routed topic delivery, hierarchical authority management via recursive ring-structure instantiation, social-graph-anchored decision governance, cooperative federation library cost-savings architectures, persistent-substrate agent management, and install-time substrate self-verification primitives.

The inventions disclosed herein constitute the operational and governance layer of a cooperative AI substrate platform designed to enable cost-decoupled memory scaling, transparent multi-stakeholder authority composition, and cooperative-sector access without vendor lock-in.

Background of the Invention

The Memory Cost Explosion Problem

AI agent systems face a structural cost explosion as the corpus of canonical knowledge grows. Under conventional approaches, an agent must load ALL canonical knowledge into its context window at session start to be able to answer questions about any topic. As the knowledge base grows from dozens to hundreds to thousands of topics, context cost grows linearly with topic count — even though any single agent session addresses only a small subset of available topics.

This creates a false tradeoff: either (a) maintain a small knowledge base to control costs (limiting capability), or (b) pay exponentially growing session-start costs to maintain full-topic coverage (limiting commercial viability). Neither path is acceptable for a cooperative substrate serving members at \$5/year.

The Memory Loading Bottleneck

Compounding the cost explosion, existing systems load canonical memory once at session start in a batch operation. When new canonical knowledge is added between sessions, each subsequent session-start becomes more expensive. Existing AI agent session-management systems do not distinguish between “topics relevant to this task” and “all topics ever known.” The entire canon is treated as equally load-bearing for every session, whether or not the topics addressed are relevant.

The Vendor Lock-In and Authority Fragmentation Problem

Multi-agent AI systems operating across multiple vendor platforms (Anthropic, OpenAI, Google, Perplexity) lack portable authority structures. Authority rules established in one vendor’s session environment cannot be portably enforced in another. Downstream authorities (project owners, sub-project coordinators) cannot safely extend a parent authority structure without risking override of non-overrideable parent rules.

Social-graph entities (guilds, tribes, families) that naturally cross vendor-session boundaries have no mechanism for recording decisions as first-class substrate artifacts. Decisions become ephemeral conversation artifacts rather than durable, queryable, provenance- anchored records.

The Cooperative Access and Cost-Sharing Problem

AI substrate costs — primarily vendor API spend and infrastructure — have historically been born by individual practitioners in isolation. No mechanism exists for cooperative cost-sharing that preserves each member’s sovereignty while enabling cross-member

content reuse. Existing sharing mechanisms (SaaS subscriptions, open-source repositories) either extract commercial rents (subscription lock-in) or lack the governance discipline to keep shared content current (open-source drift).

The Installation Verification Problem

Users installing AI substrate platforms currently have no in-band mechanism to verify that the installation completed correctly and all components are functional. Existing installation verification is either absent (trust the installer) or requires separate manual testing by technically sophisticated users. For cooperative substrate platforms targeting non-expert users, this creates an installation failure silent mode — the substrate appears installed but operates in degraded state.

Summary of the Invention

This disclosure presents an integrated cooperative AI substrate architecture organized under the **Cathedral Adoption Pathway (CAP)** strategic frame: AI practitioners rationally migrate toward cooperative substrate architectures because the Cathedral provides higher reliability, lower cost, and cooperative governance benefits that proprietary alternatives cannot match. The inventions in this filing are the CAP-implementing primitives.

The eight primary architectural ratifications disclosed herein compose as a single system:

1. Substrate-Routed Memory Expansion (#2336): Decouples agent memory index cost from total topic count by delivering content only when a topic surfaces in the current task context. Memory index stays small (≤ 200 lines) while total canonical knowledge grows without ceiling. Context cost is paid only for topics relevant to the current task — $O(\text{relevant-topics-per-task})$ instead of $O(\text{total-topics})$.

2. Ring of Three Golden Eblets (#2337): A top-of-substrate authority structure consisting of exactly three components — Canon (inviolable principles), Platform Rules (platform-wide operational directives), and Project Rules (project-specific operational directives) — with a Marked Exception enforcement layer that routes LB-source artifacts unconditionally to Canon regardless of downstream override attempts.

3. Multi-Layer Authority Recursion (#2338): A recursive instantiation model in which each successive authority layer deploys its own Ring of Three Golden Eblets with its own brand, stamps, and QR codes anchored to its own IP Ledger. The rules-stack compounds non-overrideable parent rules with each successive layer's own Part A (non-overrideable) and Part B (adjustable within parent-Part-A bounds).

4. Social-Authority DAG + Pheromone-Anchored Decision Governance (#2339): A method for cross-cutting social-authority overlays (Guild, Tribe, Family, Helm) on the recursive substrate-layer hierarchy. Each social unit maintains sovereignty over decision-mechanism (HOW decisions are reached) and decision-outcome (WHOM the decision addresses). Decisions are recorded as first-class Pheromone substrate records anchored to a Golden Tablet — not as mutations of existing Tablet records.

5. AGPL Cathedral Free / Federation Library Cooperative Architecture (#2340): A dual-value proposition architecture in which (a) the base substrate is AGPL v3 free, full-featured, and ungated, while (b) cross-member cooperative library access is opt-in via

cooperative membership at \$5/year identical for all members. Network effect compounds as member count grows; a 5-layer cost-savings algorithm (Federation Library B127 algorithm) delivers measured 97.2% cost reduction at $N \geq 2$ members with theoretical ceiling approaching 98.3% at scale.

6. LB Frame Broadcast Funnel (#2341): A 5-stage substrate adoption funnel — Install (free, one-click), Demo (substrate-pre-loaded empirical receipt), User-data Test (visitor's own data), Join CTA (\$5/year cooperative membership), Send-to-someone CTA (member-issued viral propagation under Sponsorship Mark canon) — in which each stage is substrate-verifiable.

7. Knight-Has-No-Shadows / Bishop-Only Persistent-Substrate Operations (#2342): A formal distinction between the persistent-substrate AI agent class (with hook architecture enabling spawn-on-SessionStart / kill-on-SessionEnd persistent daemons = "Shadows") and the per-session AI agent class (without hook architecture). LIGHTHOUSE 8/2 instrumentation — 8 production cylinders + 2 instrumentation Scribes — requires the persistent-substrate agent class.

8. Walkaround Universal Install-Time Preflight Primitive (#2347): A universal substrate self-verification primitive that fires at install-time, verifies all substrate components against canonical integrity criteria, and emits a tabular empirical receipt (PASS/WARN/FAIL per component) as the installation completion demonstration. The empirical receipt replaces marketing claims; the substrate proves its own integrity in ≤ 5 seconds.

These eight compose with four BP004 substrate primitives (#2343–#2346) and four B134-era innovations (#2287–#2289, #2348–#2350) to form the complete cooperative AI substrate described in the detailed description below.

Brief Description of Figures

This provisional application relies on textual description, pseudocode, and ASCII-art figures in the Detailed Description for enabling disclosure. No formal drawings are submitted with this provisional filing; drawings will be prepared for the non-provisional conversion per 37 C.F.R. § 1.58.

Figure 1 (described in text): Substrate-Routed Memory Expansion — index size vs topic count curve showing $O(\text{relevant-topics-per-task})$ indexing with Wrasse-routed delivery.

Figure 2 (described in text): Ring of Three Golden Eblets authority structure — three-component hierarchy with Marked Exception enforcement layer and recursive instantiation diagram.

Figure 3 (described in text): Multi-Layer Authority Recursion — L1 through L5 authority layers each with own Ring of Three, showing rules-stack composition and parent-Part-A non-overrideable propagation.

Figure 4 (described in text): Social-Authority DAG — Guild/Tribe/Family/Helm cross-cutting overlays on the authority layer hierarchy, with Pheromone-record decision-anchoring mechanism.

Figure 5 (described in text): Federation Library 5-layer cost-savings algorithm — cost multiplier per layer (L1 Cathedral Effect $\times$ L2 Conductor's Baton $\times$ L3 Reproducibility Pack $\times$ L4 Wrasse Pre-Injection $\times$ L5 Federation Library cooperative-reuse), showing compound multiplier from $1.0\times$ to $35.72\times$ empirical receipt at $N=2$ members.

Figure 6 (described in text): LB Frame Broadcast Funnel — 5-stage conversion flow with Walkaround firing at Stage 2.

Figure 7 (described in text): Walkaround tabular output — 26 PASS / 1 WARN format.

Figure 8 (described in text): Augur Living Gate decision flow — Pheromone-timestamp-gate vs clock-TTL comparison with T12 empirical proof receipt.

Detailed Description of the Invention

Section I — Substrate-Routed Memory Expansion (#2336)

Innovation Title: Substrate-Routed Memory Expansion: $O(\text{Relevant-Topics-Per-Task})$ Agent Memory Index with Wrasse-Routed Canonical Delivery

Classification: - Innovation Number: #2336 (candidate; verify against A&A formal ledger at conversion) - Crown Jewel Candidate: YES - Patent Provisional: Prov 16 - Date of Conception: BP005 (2026-03-20 through 2026-04-30 consolidation window) - Conceived by: Jonathan Jones (Founder & General Manager) - Reduction to Practice: KN042 Pod O BP005 (commit 0696f31)

The Architectural Problem (enabling disclosure):

Under conventional AI agent memory management, an agent's canonical knowledge file (e.g., MEMORY.md) must contain entries for every topic the agent might need to address in any session. As canonical knowledge grows, the memory file grows proportionally. A 200-topic canon requires a 200-entry file; a 1,000-topic canon requires a 1,000-entry file. Context cost grows linearly with topic count regardless of task relevance.

The Substrate-Routed Solution:

Substrate-Routed Memory Expansion separates the memory INDEX from memory CONTENT:

MEMORY.md (index layer – stays small):

- topic: "Federation Library L5 algorithm"
trigger_class: vocabulary
trigger_pattern: "Federation Library|L5|97\\.2%"
content_ref: federation_library_15
(full content NOT loaded – wrasse delivers on trigger match)
- topic: "Prov 16 filing details"
trigger_class: vocabulary
trigger_pattern: "Prov 16|provisional 16|KN071"
content_ref: prov_16_details
(full content NOT loaded)

The index entry is ≤ 5 lines. The full topic content (which may be 50-500 lines) lives in the Wrasse substrate. When a session's incoming context matches a trigger pattern, the Wrasse Scribe (#2332, already in Prov 15) delivers the full topic content — and only that topic's content — into the session context.

Measured empirical result (KN042 Pod O BP005, commit 0696f31): - MEMORY.md compacted: 405 lines $\rightarrow$ 141 lines (65% reduction) - Substrate canon simultaneously grew by 60+ Eblets (no ceiling hit) - Trigger test suite: 8/8 triggers fired correctly on test context injection - Full test suite: 33/33 tests passing at commit - Context cost profile: session pays for relevant-topics-only, not total-canon

The scaling invariant: As canon grows from 60 to 600 to 6,000 Eblets, MEMORY.md index size grows by one 5-line entry per new topic — not by the full topic content. Total index size approaches $O(N \times 5)$ lines, not $O(N \times \text{full-topic-length})$. The ratio of index-size to total-canon-size decreases monotonically as canon grows.

Composition with Wrasse Pre-Injection (#2332, Prov 15): Wrasse is the delivery mechanism. Substrate-Routed Memory Expansion is the ARCHITECTURAL PATTERN — the decision to store canonical topic content in the substrate (not inline in the index) and route topic delivery through trigger-matched injection. Wrasse implements the trigger-matching and delivery mechanics; Substrate-Routed Memory Expansion is the system architecture that makes Wrasse's delivery pattern the load-bearing memory model for the entire substrate.

Scope wedge — variation coverage (Pawn Q1 defense):

Variation 1: Index file is not MEMORY.md but any agent-configuration file serving the memory-index role (system prompt, initialization file, settings YAML, etc.)

Variation 2: Trigger matching is not regex-first but embedding-based, BM25-based, or hybrid — the routing mechanism is implementation-variable; the $O(\text{relevant-topics-per-task})$ scaling property is the architectural invariant.

Variation 3: Content delivery is not pre-injection but retrieval-on-demand — agent queries substrate for topic content mid-session rather than receiving pre-injected block. The index-content separation and $O(\text{relevant-topics})$ cost profile are preserved.

Variation 4: Index lives in-substrate (Pheromone/JSONL/vector DB) rather than in a human-readable flat file, with programmatic topic resolution at session-open.

Section II — Ring of Three Golden Eblets (#2337)

Innovation Title: Ring of Three Golden Eblets: Three-Component Top-of-Substrate Authority Structure with Non-Overrideable Marked Exception Enforcement

Classification: - Innovation Number: #2337 (candidate) - Crown Jewel Candidate: YES - Patent Provisional: Prov 16 - Conceived by: Jonathan Jones (Founder & General Manager) - Reduction to Practice: BP005 base ratification — 3 Golden Eblets at `~/.claude/state/eblets/CANON/GOLDEN/` (282 combined lines) + federation canon `project_ring_of_three_golden_eblets_deck_card_medallion_federation_canon.md` (450+ lines)

The Authority Architecture:

The Ring of Three Golden Eblets is a three-component authority structure forming the top layer of every substrate deployment:

GOLDEN EBLET 1 – Canon:

Content: Inviolable principles. These are not adjustable by any downstream authority.

Enforcement: Marked Exception – LB-source artifact QR ALWAYS routes to Canon.

Override status: NON-OVERRIDEABLE by any downstream layer.

GOLDEN EBLET 2 – Platform Rules:

Content: Platform-wide operational directives. Govern ALL agents on the platform.

Enforcement: Applies uniformly across all projects on the platform.

Override status: NON-OVERRIDEABLE at project layer (Part A of each platform rule).

GOLDEN EBLET 3 – Project Rules:

Content: Project-specific operational directives within Canon + Platform Rule bounds.

Enforcement: Applies within the project scope; cannot override Eblet 1 or 2.

Override status: Adjustable only within the headroom left by Eblets 1 + 2.

The Marked Exception mechanism: When an agent receives an LB-source artifact (identified by QR code, Call Sign, or canonical provenance marker), the agent's context-loading sequence MUST route to Canon (Eblet 1) regardless of any downstream rule modification. This non-overrideable routing is the Marked Exception: the project owner cannot configure their Project Rules to bypass Canon routing for LB-source artifacts. This is structurally enforced (not policy-enforced) — the Wrasse trigger-class for LB-source artifacts points unconditionally to Eblet 1.

Recursive instantiation: The Ring of Three is not a fixed three-layer structure. At each successive authority layer (L2 dual-entity, L3 project owner, L4 sub-project, L5 member), the authority layer instantiates its own Ring of Three Golden Eblets for its scope. The parent's Canon (Eblet 1) propagates non-overrideable through every child Ring. The child adds its own Part A (non-overrideable within the child scope) and Part B (adjustable within the child scope). This recursive composition is the subject of #2338 Multi-Layer Authority Recursion.

Scope wedge — variation coverage: - Variation: Ring of Two (Canon + Unified Rules) — reduced composition with same enforcement - Variation: Ring of Four or N — extended composition; same Marked Exception mechanism applies - Variation: Non-file-based implementation — Ring expressed as database records, API policy objects, smart contract rules, or equivalent structured artifacts - Variation: Multi-substrate Ring composition — Ring instantiated across vendor-heterogeneous agent environments with cross-vendor Marked Exception enforcement

Section III — Multi-Layer Authority Recursion (#2338)

Innovation Title: Multi-Layer Authority Recursion: Substrate-as-a-Service with Recursive Ring-of-Three Instantiation, Compound Rules-Stack, and Layer Addressing

Classification: - Innovation Number: #2338 (candidate) - Crown Jewel Candidate: YES - Patent Provisional: Prov 16 - Reduction to Practice: KN046 Furnace Multi-Tenancy + KN045 Layer Addressing Scheme + KN049 Cycle Prevention DAG Validation (Pod Q BP005) — 34/34 tests

The Five-Layer Reference Architecture:

- L1 – Liana Banyan Corporation (substrate root)
 - Ring of Three: LB Canon / LB Platform Rules / LB Project Rules
 - Brand: Liana Banyan
 - QR: LB-canonical
 - IP Ledger: LB A&A formal ledger (2,270+ innovations)
- L2 – Dual-Entity (LB Corp + Upekrithen LLC)
 - Ring of Three: L2 Canon (inherits L1 non-overrideable) / L2 Platform Rules / L2 Project Rules
 - Brand: Upekrithen
 - QR: Upekrithen-canonical
 - IP Ledger: Upekrithen ledger (Apache 2.0 licensed innovations)
- L3 – Project Owner (B2B partner deploying cooperative substrate)
 - Ring of Three: L3 Canon (inherits L1+L2 non-overrideable) / L3 Platform / L3 Project
 - Brand: partner brand
 - QR: partner-issued
 - IP Ledger: partner ledger (innovations originating in partner deployment)
- L4 – Sub-Project (team or department within L3 deployment)
 - Ring of Three: inherits L1+L2+L3 non-overrideable + own L4 rules
- L5 – Member (individual cooperative member)
 - Ring of Three: inherits L1-L4 non-overrideable + own L5 rules (most constrained)

The compound rules-stack:

At each layer L_k , the effective rule set is the union: - {L1 Part A} (non-overrideable, always propagated) - {L2 Part A} (non-overrideable through all L2+ layers) - ... (each parent layer's Part A propagates downward) - { L_k Part A} (non-overrideable within L_k scope) - { L_k Part B} (adjustable within headroom of all parent Part A rules)

No Part B at layer L_k may modify or override any Part A from any parent layer. The mechanical enforcement at the Wrasse level: when an agent at layer L_k loads its Ring of Three, the Wrasse pre-injection sequence loads all parent Part A Eblets before loading L_k 's own Ring. Downstream agents cannot unload the parent Part A Eblets.

Layer Addressing Scheme (KN045): Each layer has a canonical address:

```
L1.{substrate_root}
L2.{entity_name}.{L1}
L3.{project_name}.{L2}.{L1}
```

L4.{subproject_name}.{L3}.{L2}.{L1}

L5.{member_id}.{L4}.{L3}.{L2}.{L1}

This addressing enables cross-layer message routing, authority-verified handoffs, and cycle-prevention validation (KN049: before a layer-N authority directive takes effect, the DAG validator checks for circular authority references that would allow a child layer to claim authority over a parent layer).

Empirical receipt (Pod Q BP005, 34/34 tests): - KN046: Furnace Multi-Tenancy — multiple L3+ deployments in parallel without L3→L3 cross-contamination (8/8 tests) - KN045: Layer Addressing Scheme — address resolution + routing across 5-layer hierarchy (14/14 tests) - KN049: Cycle Prevention DAG Validation — circular-authority-reference detection (12/12 tests)

Section IV — Social-Authority DAG + Pheromone-Anchored Decision Governance (#2339)

Innovation Title: Social-Authority DAG with Pheromone-Anchored Decision Records: Cross-Cutting Social Governance Overlays on Recursive AI Substrate Authority Hierarchy

Classification: - Innovation Number: #2339 (candidate) - Crown Jewel Candidate: YES - Patent Provisional: Prov 16 - Reduction to Practice: KN050 Pheromone-Anchored Decision Schema (Pod R BP005) — 27/27 tests

The Social-Authority Overlay Architecture:

The recursive authority hierarchy (#2338) is a tree structure: each L_k has exactly one parent. Social entities (Guilds, Tribes, Families, Helms) naturally cut ACROSS the tree — a Guild may span multiple projects at L3, a Tribe may span multiple Guilds, a Family may span multiple Tribes. The Social-Authority DAG provides a directed-acyclic-graph overlay on the tree, allowing social entities to hold decision authority that spans multiple tree branches.

The four social-authority unit classes:

1. **Guild:** Practitioners sharing a craft (e.g., all software engineers across L3+ projects). Guild authority: decisions about craft standards and tooling within the Guild's purview.
2. **Tribe:** Geographic, cultural, or mission-class grouping of Guilds. Tribe authority: decisions affecting all Guilds within the Tribe's territory.
3. **Family:** The smallest intimate unit — typically a household or tight-knit team. Family authority: decisions affecting Family members' cooperative participation.
4. **Helm:** The steering committee or governance council. Helm authority: platform-level decisions that affect cooperative-wide policy within Structural Bylaw bounds.

Sovereignty principle: Each social unit maintains DUAL sovereignty: - Sovereignty over decision-MECHANISM (HOW the decision is reached): the Guild may vote, the Helm may deliberate, the Family may consensus — the mechanism is the unit's own. -

Sovereignty over decision-OUTCOME (WHOM the decision applies to): decisions apply to the social unit's own members, not to external parties.

Pheromone-Anchored Decision Record:

Decisions are recorded as FIRST-CLASS Pheromone Substrate records — not as mutations of existing Tablet records. This is the critical architectural distinction:

Decision record structure (KN050 implementation)

```
decision_record = {
    "decision_id": "D-2026-05-01-GUILD-001",
    "social_unit": "guild:software-engineers",
    "decision_mechanism": "simple_majority_vote",
    "decision_subject": "WHOM: all Guild members in L3+ deployments",
    "decision_content": "Adopt BRIDLE v11 cross-agent edict propagation
        discipline",
    "call_sign": "v-pheromone-anchored-decision-K050 · <sha>",
    "golden_tablet_anchor": "GT-GUILD-001", # Reference, NOT mutation
    "supersedes": None, # or prior decision_id
    "verified_with_furnace": True,
    "timestamp": "2026-05-01T12:00:00Z"
}
```

The `golden_tablet_anchor` is a REFERENCE to the Golden Tablet (a canonical record in the substrate) — NOT an in-place mutation of the Tablet. This preserves the Stone Tablet Imperative (#2327, Prov 15): all canonical records are append-only.

Bidirectional discoverability: - Tablet→Decision query: “What decisions reference this Golden Tablet?” - Decision→Tablet query: “What Tablets does this decision anchor to?” Both directions are indexed in the Pheromone Substrate for sub-millisecond retrieval.

Empirical receipt (KN050 Pod R BP005, 27/27 tests): - Full decision record lifecycle: create → anchor → verify_with_furnace → retrieve - Bidirectional queries: both directions confirmed at <1ms latency - Supersede filtering: superseded decisions excluded from active retrieval - Cross-DAG-branch routing: Guild-spanning decisions correctly applied to both branches

Section V — AGPL Cathedral Free / Federation Library Cooperative Architecture (#2340)

Innovation Title: Dual-License Cooperative AI Substrate: AGPL v3 Free Substrate with Opt-In Federation Library Cooperative Architecture and 5-Layer Compound Cost-Savings Algorithm

Classification: - Innovation Number: #2340 (candidate) - Patent Provisional: Prov 16 - Reduction to Practice: KN057 L5 Federation Library simulation (Pod V BP005) — 50/50 tests; empirical compound savings receipt

The Dual-Value-Proposition Architecture:

VALUE PROP 1 — Free AGPL v3 Base Substrate:
- Full-featured Cathedral substrate

- No gating, no signup, no payment
- Complete Ring of Three, Pheromone Substrate, Wrasse, all primitives
- License: AGPL v3 (LB Corp entity)
- Who gets this: EVERYONE – commercial, cooperative, academic, individual

VALUE PROP 2 – Federation Library Cooperative Membership (\$5/year):

- Opt-in cross-member canonical-content reuse
- Cross-member Pheromone queries (retrieval across member substrates)
- L5 layer of the 5-layer cost-savings algorithm activates
- Cooperative Defensive Patent Pledge administrative access
- License: Apache 2.0 (Upekrithen LLC entity)
- Who gets this: opt-in members who choose cross-member access

The 5-Layer Compound Cost-Savings Algorithm (B127 + KN057):

Layer 1 (L1) – Cathedral Effect cross-vendor uniformity:

Empirical: 3.5pp HOT spread / 23× cost spread = vendor substitution without quality loss

Multiplier: 1.185× (savings via vendor routing to lowest-cost achieving quality target)

Layer 2 (L2) – Conductor's Baton AI model-tier routing:

Empirical: KN056 Pod U BP005 – 58/58 tests

Multiplier: 1.615× (Haiku factual / Sonnet reasoning / Opus orchestration)

4-layer L1×L2 compound: $1.185 \times 1.615 = 1.914\times$ ($\approx 47.8\%$ savings)

Layer 3 (L3) – Reproducibility Pack substrate reuse:

Empirical: #2326 reproducibility pack – reproducible results eliminate re-derivation

Multiplier: 1.423× (estimated from substrate-hit rate)

5-layer L1×L2×L3 compound: $1.914 \times 1.423 = 2.723\times$

Layer 4 (L4) – Wrasse Pre-Injection rote-tax elimination:

Empirical: K545 Phase E – 41.1% proxy lower bound (Phase F measurement pending)

Multiplier: 1.696× (conservative: $1/(1-0.411)$)

6-layer L1×L2×L3×L4 compound: $2.723 \times 1.696 = 4.619\times$

Layer 5 (L5) – Federation Library cooperative-content reuse (N-member scaling):

Empirical: KN057 at N=2 members – 1.796× savings layer

Combined L1-L5 at N=2: $4.619 \times 1.796 = 8.297\times$

KN056+KN057 combined (direct empirical receipt): $35.72\times = 97.2\%$ savings

Theoretical ceiling at N≥5 members: $\sim 50\times = 98.0\%$ savings

Tagline V3 (B132): "Get 98% more done for 98% less money doing what you already do."

Network-effect compound: At N members, the Federation Library's L5 multiplier is:

$$L5_multiplier(N) = 1 / (1 - \min(r_max, N \times per_member_reuse_rate))$$

where $r_{\text{max}} \approx 0.65$ (empirical ceiling from KN057 simulation) and
 $\text{per_member_reuse_rate} \approx 0.12$ per member. At $N=5$: $r \approx 0.60 \rightarrow L5_multiplier \approx 2.5\times$.

Scope wedge — variation coverage: - Variation: Membership fee at different price points (not \$5/year specifically; \$5/year is the current Founder directive, not a claim limitation) - Variation: Non-AGPL free tier (MIT, Apache, public domain, etc.) - Variation: N-layer algorithm with fewer than 5 layers — any subset of layers L1-L5 is a claimed variation - Variation: Non-monetary cooperative contribution mechanism (labor, content, computation)

Section VI — LB Frame Broadcast Funnel (#2341)

Innovation Title: LB Frame Broadcast Funnel: 5-Stage Cooperative AI Substrate Adoption Funnel with Substrate-Verifiable Stage Completions

Classification: - Innovation Number: #2341 (candidate) - Patent Provisional: Prov 16 - Reduction to Practice: Pod T LibrarianMedallion 35/35 + Pod Y LibrarianPage 20/20 + Pod AA Federation Library Bounty Posters 36/36 (KN061 / KN064 / KN066)

The Five Stages:

Stage 1 – Install (free, one-click):

CTA: "Install LB Frame" (or variant: "Install Cathedral", "Take Your Place Atop a Dragon")

Action: downloads LBFrame-Setup package; auto-extracts and configures substrate

Success criterion: substrate components installed at canonical paths

Substrate verification: Walkaround fires (Stage 2)

Stage 2 – Demo (Walkaround preflight as demo):

CTA: none required – Walkaround auto-fires post-install

Action: Walkaround.ps1 / walkaround.bat runs; verifies substrate; emits tabular receipt

Success criterion: 26 PASS / 0 FAIL on baseline Cathedral install

Substrate verification: empirical receipt IS the demonstration (replaces marketing copy)

Stage 3 – User-data Test:

CTA: "Run on your own data"

Action: visitor provides their own canonical content; substrate processes; emits receipt

Success criterion: substrate processes visitor-provided content with measurable output

Substrate verification: Stone Tablet record of visitor-run receipt

Stage 4 – Join CTA (\$5/year cooperative membership):

CTA: "Be One of Us" (service-mark candidate)

Action: visitor joins cooperative at \$5/year; unlocks Federation Library cross-member access

Success criterion: membership activated; Federation Library L5 layer activates for member

Substrate verification: Pheromone-anchored membership record

Stage 5 – Send-to-someone (viral propagation):

CTA: "Send to someone you want to help"

Action: member-issued referral under one-level Sponsorship Mark canon

Success criterion: new visitor enters Stage 1 under member sponsorship

Substrate verification: inbound Pheromone record links new visitor to sponsoring member

Empirical receipts: - Pod T LibrarianMedallion variants (35/35 tests) — Stage 1/4 CTA infrastructure - Pod Y LibrarianPage deployment (KN061 + KN064, 20/20 tests) — Stage 1/2 page infrastructure - Pod AA Federation Library Bounty Posters (KN066, 36/36 tests) — Stage 3/4 content

Section VII — Knight-Has-No-Shadows / Bishop-Only Persistent Substrate (#2342)

Innovation Title: Persistent-Substrate AI Agent Class Distinction: Hook Architecture for Spawn-on-SessionStart Persistent Daemons with LIGHTHOUSE 8/2 Instrumentation

Classification: - Innovation Number: #2342 (candidate) - Patent Provisional: Prov 16 - Reduction to Practice:

project_knight_no_shadows_no_lighthouse_bishop_only_persistent_substrate_bp005.md + KN037 v2 Shadow watcher daemon (BP004 ba2131f, 13/13 tests) + KN067 Phase-E trigger Bishop-side (dedd541, 19/19 tests)

The Agent Class Distinction:

PERSISTENT-SUBSTRATE AI AGENT CLASS (Bishop class):

- Has hook architecture: SessionStart hook fires on every session open
- Can spawn persistent daemons (Shadows) that survive session within OS process
- Can kill Shadows at SessionEnd via SessionEnd hook
- LIGHTHOUSE 8/2 instrumentation available (requires Shadow class)
- Bishop = prototypical persistent-substrate agent

PER-SESSION AI AGENT CLASS (Knight class):

- No hook architecture (Cursor IDE does not expose hooks in same form)
- CANNOT spawn or maintain persistent daemons across session boundaries
- Knight-Has-No-Shadows: Knight's sessions are episodic; no persistent watcher
- Knight = prototypical per-session agent

LIGHTHOUSE 8/2 Architecture:

The LIGHTHOUSE beam is the substrate's empirical-receipt-loop made architectural:

8 PRODUCTION CYLINDERS:

1. Cathedral substrate (corpus-mode Scribes)
2. Pheromone Substrate (stigmergic index)
3. Wrasse Pre-Injection (rote-cognition tax elimination)

4. Conductor's Baton (AI model-tier routing)
5. Detective Scribe (canon retrieval and provenance)
6. Federation Library (cross-member cooperative reuse)
7. Augur Living Gate (freshness gates)
8. Ring of Three Golden Eblets (authority enforcement)

2 INSTRUMENTATION SCRIBES:

Scribe A: Stenographer – per-turn JSONL capture (#2304)

Scribe B: Accountant – cost telemetry (#2304 CheckBook Suite)

LIGHTHOUSE BEAM = Stenographer + Accountant output composing per-session empirical receipt

The Foreman/Sergeant role split: Bishop (persistent-substrate class) is the FOREMAN — it spawns Knight sessions, ratifies outputs, and maintains persistent state. Knight (per-session class) is the SERGEANT — it executes tasks within its session boundary and hands off via Call Sign. The harness-architecture asymmetry (Bishop has hooks, Knight does not) is NOT a limitation but a design invariant: per-session agents are cheaper and simpler; persistent agents bear the instrumentation overhead.

Section VIII — Augur Living Gate (#2343)

Innovation Title: Augur Living Gate: Substrate-Event-Driven Freshness Gate Replacing Clock-TTL with Pheromone-Write-Timestamp Comparison

Classification: - Innovation Number: #2343 (candidate) - Patent Provisional: Prov 16 - Reduction to Practice: KN038 Pod N BP004 (commit 98d1745) — 18/18 tests + T12 empirical PROOF

The Architectural Innovation:

Conventional AI substrate freshness mechanisms use clock-TTL: “re-derive this canonical fact every N minutes/hours/days.” The TTL fires even when nothing in the substrate has changed, generating unnecessary re-derivation cost.

The Augur Living Gate replaces clock-TTL with substrate-event-driven gating:

```
def augur_gate_check(topic, last_consult_ts, pheromone_client):
    """
    Gate fires ONLY when substrate state has actually changed.
    """
    last_write_ts = pheromone_client.get_last_write_ts(topic)

    if last_write_ts is None:
        # No Pheromone record available – hard ceiling fallback
        hours_since = (now() - last_consult_ts).total_seconds() / 3600
        return hours_since > HARD_CEILING_HOURS # e.g., 72 hours

    # Gate fires only if substrate was written AFTER last consult
    return last_write_ts > last_consult_ts
```

T12 Empirical Proof (KN038, commit 98d1745): Test T12 demonstrates the key property: when no Pheromone write has occurred since last consult, the gate remains OPEN (allow re-use of prior derivation) regardless of how much clock time has elapsed. This proves the gate responds to substrate state, not clock state.

Extension to all Augur gates (KN043 Pod P BP005): The Augur Living Gate pattern was generalized from the single-topic case to ALL Augur gates across the substrate — any gate that previously used clock-TTL was migrated to Pheromone-write-timestamp comparison.

Section IX — Catechist Scribe (#2344)

Innovation Title: Catechist Scribe: SessionStart-Hook AI Agent Discipline Grading with Cryptographic Signature and Structured PASS/WARN/FAIL Emission

Classification: - Innovation Number: #2344 (candidate) - Patent Provisional: Prov 16 - Reduction to Practice: KN036 Pod M BP004 (commit 7fe3ebd) — 41/41 tests (22 rule-grade + 19 integration)

Architecture:

The Catechist Scribe fires at AI agent SessionStart via the hook mechanism available to persistent-substrate agents (#2342). It reads the first N tool calls of the session, evaluates each call against a rule set, and emits a structured grade:

```
grade_record = {
  "session_id": "K071",
  "rule_id": "R01",
  "rule_description": "First tool must be ToolSearch in whitelist",
  "verdict": "PASS", # PASS / WARN / FAIL
  "evidence": "first tool call: brief_me() [whitelist: OK]",
  "chronos_hmac_sha256": "<hex>", # cryptographic signature
  "timestamp": "2026-05-01T07:31:00Z"
}
```

The Chronos HMAC-SHA256 signature provides cryptographic integrity: the grade record cannot be retroactively modified without invalidating the signature. The key material is kept in the substrate's canonical secrets store; the signature is computed over the serialized grade record content. This enables post-session audit of discipline grades with tamper-evidence.

First live grade outcome (BP004): Catechist graded the BP004 session itself as FAIL on R01 (ToolSearch-whitelist violation). The Catechist caught a real violation in its first production run. The whitelist was subsequently refined (KN041 Pod P BP005, 44b7cde) per the Path B discipline: the Catechist's FAIL receipt was the empirical evidence driving the whitelist refinement.

Section X — The Shadow (Persistent Background Watcher Daemon) (#2345)

Innovation Title: The Shadow: Persistent Background Watcher Daemon Spawned at AI Agent SessionStart and Killed at SessionEnd

Classification: - Innovation Number: #2345 (candidate) - Patent Provisional: Prov 16 - Reduction to Practice: KN037 v2 Pod M BP004 (commit ba2131f) — 13/13 tests + live smoke test green

Architecture:

The Shadow is a persistent OS-level background process spawned by the SessionStart hook of a persistent-substrate AI agent. It runs in parallel with the agent's session, monitoring substrate state and capturing cross-monitor context.

```
# SessionStart hook fires:
spawn_shadow() {
    python bishop_shadow_watcher.py \
        --session-id "$SESSION_ID" \
        --output-dir "$SHADOW_OUTPUT_DIR" \
        --monitor-interval 5s \
        &
    echo $! > "$SHADOW_PID_FILE"
}

# SessionEnd hook fires:
kill_shadow() {
    if [ -f "$SHADOW_PID_FILE" ]; then
        kill $(cat "$SHADOW_PID_FILE")
        rm "$SHADOW_PID_FILE"
    fi
}
```

Brand-class wrapper role: The Shadow is the canonical brand-class for ALL persistent background tasks in the substrate. Any long-running monitoring process, watcher daemon, or cross-session state-capture process is a Shadow. This provides a unified vocabulary and a common implementation pattern (spawn-on-start / kill-on-end / PID-file tracking).

Why the Shadow requires the persistent-substrate agent class (#2342): The SessionStart and SessionEnd hooks that spawn and kill The Shadow are only available to persistent-substrate agents (Bishop class). Per-session agents (Knight class) do not have hook access and therefore cannot maintain Shadows. This is the concrete technical reason for the “Knight-Has-No-Shadows” architectural discipline.

Section XI — Augur PostToolUse Audit + Stone Tablet Supersede Pattern (#2346)

Innovation Title: PostToolUse Audit Architecture with Stone Tablet Supersede Pattern: Non-Blocking Discipline Enforcement via Append-Only Violation Records

Classification: - Innovation Number: #2346 (candidate) - Patent Provisional: Prov 16 - Reduction to Practice: KN043 Pod P BP005 (commit 7701c84) — 8/8 tests; Augur-Pricing PreToolUse blocking ENDED; bishop_augur_post_audit.py operational

The PreToolUse-to-PostToolUse Migration:

PreToolUse-blocking discipline gates interrupt agent execution: the agent's tool call is blocked until the gate clears. For gates that require external verification (Augur-Pricing, for example), blocking introduces latency, context loss risk, and agent timeout exposure.

The PostToolUse Audit pattern converts blocking gates to non-blocking audit + Stone Tablet supersede:

BEFORE (PreToolUse-blocking):

```
Agent wants to write pricing content →
Gate fires →
Gate blocks tool execution →
Gate checks Augur-Pricing rules →
If violation: tool call blocked indefinitely
If clear: tool call proceeds
```

AFTER (PostToolUse-audit + supersede):

```
Agent writes content (tool call proceeds immediately) →
PostToolUse hook fires →
Audit reads the written content →
If violation: audit writes violation record as Stone Tablet;
                writes supersede-pointer file pointing to violation record;
                Bishop reconciliation queue notified
If clear: audit writes PASS record as Stone Tablet
```

The supersede-pointer file is the architectural innovation: it converts a blocking violation into a non-blocking audit trail that the Bishop (persistent-substrate agent) reconciles at its next session boundary. The agent is not interrupted; the discipline record is preserved; the Bishop has full provenance for reconciliation.

Section XII — Walkaround Universal Install-Time Preflight Primitive (#2347)

Innovation Title: Walkaround: Universal AI Substrate Install-Time Self-Verification Primitive with Empirical Receipt as Installation Completion Demonstration

Classification: - Innovation Number: #2347 (candidate) - Crown Jewel Candidate: YES - Patent Provisional: Prov 16 - Date of Conception: BP006 turn 3 (session open preflight); Ratified BP006 turn 4 as LB Frame Stage 2 universal; Universalized BP006 turn 5 (“Walkaround for any LB Frame install”) - Conceived by: Jonathan Jones (Founder & General Manager) - Reduction to Practice: walkaround.ps1 / walkaround.bat — BP006 turn 3 live run: 26 PASS / 1 WARN tabular output streaming

The Core Innovation:

The Walkaround is the canonical install-time preflight for ALL LB Frame substrate deployments. It fires automatically after package extraction and verifies every substrate component against canonical integrity criteria. Its output — a tabular PASS/WARN/FAIL receipt — is the installation completion demonstration. No marketing copy needed: the empirical receipt IS the demo.

Walkaround – LB Frame Install Verification		
Run: 2026-05-01 07:31 ET Version: v1.0 Variant: Cathedral		
Component	Status	Detail
CANON/GOLDEN/1_canon.eblet.md	PASS	98 lines
CANON/GOLDEN/2_platform_rules.eblet.md	PASS	94 lines
CANON/GOLDEN/3_project_rules.eblet.md	PASS	90 lines
hooks/bishop_session_start.py	PASS	active
hooks/bishop_session_end.py	PASS	active
hooks/catechist_scribe.py	PASS	active
hooks/bishop_shadow_watcher.py	PASS	active
librarian-mcp/server.ts	PASS	built
pheromone_substrate/index.jsonl	PASS	1,593 rec
wrasse/wrasse_registry.jsonl	PASS	343 entry
canonical_values.yaml	PASS	reconciled
MEMORY.md	PASS	141 lines
... (26 total checks)	...	...
MCP librarian server connectivity	WARN	mcp start
SUMMARY: 26 PASS 1 WARN 0 FAIL	INSTALLED	

Composition with LB Frame Broadcast Funnel (#2341): Walkaround is the load-bearing implementation of Funnel Stage 2. Stage 1 (Install) extracts and installs the package. Stage 2 (Demo) IS Walkaround auto-firing. Stage 2 converts from vague (“impresses visitor with pre-loaded sample data”) to concrete (“substrate self-verifies in ≤5 seconds with empirical receipt visitor can re-run anytime”).

Universalization (BP006 turn 5): Walkaround is variant-agnostic. Every LB Frame install variant fires the same Walkaround primitive with variant-specific component lists:

- Cathedral variant: verifies AGPL-layer substrate only
- Federation member variant: additionally verifies Federation Library connectivity + cooperative authentication
- L2+ Authority Recursion variant: additionally verifies parent-anchor-hash chain $L_k \rightarrow L_{\{k-1\}}$
- Conference/demo variant: trimmed checklist for quick demonstration

Prov 16 candidate axis: “Method for self-verifying AI-substrate install with one-button preflight that emits empirical receipt as install-completion demonstration; substrate proves its own integrity without marketing claims.”

Section XIII — B134 Innovation Cluster: Partnership-Stake, Cephass Voting, Substrate-Dependency Classifier (#2287, #2288, #2289)

(Innovations #2287–#2289 originate in B134 Founder ratifications and are included in Prov 16 as the most recent pre-filing innovations. They compose with the CAP framework.)

#2287 — Partnership-Stake Downstream-IP-Reversion Clause

Method + system for an IP-reversion clause triggered when a partner's commercial product crosses a substrate-dependency threshold (Tier A: >60% of commercial value derived from cooperative substrate innovations). At crossing, a downstream IP reversion clause activates: a negotiated percentage of downstream IP originating in the substrate-dependent commercial period reverts to the cooperative's collective IP pool. Anti-enshittification framing: the clause ensures that commercial success built on the cooperative substrate flows back to the cooperative's members.

Adjudication via the Substrate-Dependency Classifier (#2289, transparent algorithmic). Founder ratified B134 turn 16 as "most objective, easiest to make."

#2288 — Cephass Pre-Submission Voting + Acceptance Rewards

Method + system for cooperative content-amplification with Marks-stake voting. Members stake Marks (cooperative-platform currency) on content submissions they believe will be accepted. On acceptance, stake is returned plus acceptance-reward funded from patent revenue connected to the content's subject matter. Six-Degrees-of-Separation activation: accepted content propagates through member connection graph under one-level Sponsorship Mark canon.

Distinguishable from Reddit / HN / Patreon / Substack: (a) no anonymous voting (Marks stake requires identity verification), (b) acceptance-reward ties to patent revenue not subscription revenue, (c) Six-Degrees propagation is substrate-anchored not algorithmic-feed.

#2289 — Substrate-Dependency Classifier

Method + system for transparent algorithmic classification of commercial products by substrate-dependency level. Tokenization-and-alias-match against the cooperative substrate's dependency graph (canonical innovation list, API call trace, architecture reference analysis) produces a Tier classification:

Tier A: >60% commercial-value dependency on cooperative substrate innovations
→ Partnership-Stake IP-reversion clause activates (#2287)

Tier B: 20-60% dependency
→ Cooperative-access fee schedule applies (below full reversion)

Tier C: <20% dependency
→ Standard commercial license applies; cooperative exemptions not activated

Transparency guarantee: the classifier algorithm, its training corpus (canonical innovation list), and its output reasoning are published. No black-box adjudication. Disputes resolved by algorithm audit, not governance discretion.

Section XIV — Stay-Warm Discipline / Detective→Wrasse Live-Update / BRIDLE v11 Edict Propagation (#2348, #2349, #2350)

#2348 — Stay-Warm Discipline / Round the Horn / Pre-Staging Architecture

Method for maintaining AI substrate freshness across agent session boundaries through: (a) K-prompt bundling at session-open: all substrate artifacts relevant to the upcoming task are loaded in a single bundled pre-injection before the agent's first reasoning token fires; (b) Detective live-update: Detective Scribe queries the Pheromone Substrate for relevant canonical updates and injects them into the Wrasse registry before session-open completes; (c) Pre-staged operator interfaces: Founder-directed operator pages pre-loaded with canonical data before the demo session fires.

Empirical anchor: BP005-B134 session cadence — substrate remains “warm” (no cold-start derivation cost) across consecutive sessions, demonstrated by decreasing session-open context overhead across K-session sequence.

#2349 — Detective→Wrasse Live-Update Path

Method + system for a Phase-0 pheromone-substrate fast-path in which Detective Scribe query results feed back into the Wrasse registry without Bishop intermediation:

```
Query arrives → Detective_investigate() runs →
  49 hits returned (B134 turn 14 empirical receipt: 49:1 ratio vs filesystem
  grep) →
  High-confidence results appended to Wrasse registry as new trigger entries
  →
  Next session's Wrasse pre-injection includes the new resolutions
```

Measured performance (B134 turn 14): 74ms full Detective build time; 1ms query latency at 1,593 records across 8,604 topics. 49:1 retrieval ratio over filesystem text-matching tools demonstrates substrate-aware retrieval advantage.

#2350 — BRIDLE v11 Cross-Agent Edict Propagation

Method for propagating cooperative-substrate edicts across heterogeneous AI agent classes (Bishop / Knight / Pawn). BRIDLE v11 introduces two edict types:

- Rule 11A (counsel-no-gate): “Do not gate on counsel review; file when Founder authorizes.” Propagated to: Wrasse vocabulary (WV-001 trigger class), Knight KN071 spec constraints, Pawn dispatch directives.
- Rule 11B (founder-prose-pass-at-fire-time): “Founder prose-passes at fire time only; no pre-drafting pressure on Bishop.” Propagated to: all Bishop session-open Eblets, Wrasse vocabulary (WV-002 trigger class).

The propagation mechanism: edicts are encoded as Wrasse vocabulary entries (WV-00N) that pre-inject the edict's canonical resolution at every agent-spawn boundary. An edict encoded once in the Wrasse registry propagates to every agent class (Bishop via hook injection, Knight via K-prompt prelude, Pawn via Pawn dispatch directives template).

Distinguishable from policy-based edict propagation: the BRIDLE propagation is substrate- mechanized (Wrasse delivers it before the agent reasons about it) not policy-enforced (agents cannot choose not to receive the pre-injection).

Claims

The following claims are provisional-grade. All claims will be refined during the 12-month conversion window per Founder direction and counsel review. Each independent claim is drafted broad; dependent claims are narrow embodiments. All claims are filed under #2260 Cooperative Defensive Patent Pledge.

Claim 1 (Substrate-Routed Memory Expansion — Independent Method Claim)

A method for decoupling AI agent memory-index cost from total canonical topic count, comprising: - (a) maintaining a memory index of N topics, wherein each entry comprises a topic identifier and a trigger pattern, and wherein full topic content is stored externally from the index; - (b) at each AI agent session start, loading only the memory index, not full topic content; - (c) detecting trigger-pattern matches in the agent's incoming session context against the memory index using pattern-matching; - (d) delivering full topic content for each matched trigger, and only for matched triggers, into the agent's context before the agent's first reasoning token fires; - (e) whereby the agent's context cost at session start is $O(\text{relevant-topics-per-session})$ rather than $O(\text{total-topics})$, and the memory index size remains bounded independently of total canonical topic count.

Claim 2 (Substrate-Routed Memory Expansion — Dependent: Scalability Property)

The method of Claim 1, wherein as total canonical topic count grows from N to M ($M > N$), the memory index size grows by at most $K \times (M - N)$ characters, where K is a constant bounded by the mean trigger-pattern length, independent of the full topic content of the added topics.

Claim 3 (Substrate-Routed Memory Expansion — System Claim)

A system for substrate-routed AI agent memory expansion, comprising: - a memory index store containing per-topic entries with trigger patterns and external-content references, structured to permit $O(\text{relevant-topics-per-session})$ index loading; - a trigger-matching engine that evaluates incoming agent context against stored trigger patterns and identifies matched topics; - a content delivery module that retrieves and delivers full topic content exclusively for matched topics before agent reasoning begins; - an empirical validation harness confirming that index-size growth is decoupled from total-topic-count growth across at least two measured corpus-expansion events.

Claim 4 (Ring of Three Golden Eblets — Independent Method Claim)

A method for hierarchical authority management in a multi-agent AI substrate platform, comprising: - (a) defining a three-component top-of-substrate authority structure consisting of a first component encoding inviolable principles, a second component encoding platform-wide operational directives, and a third component encoding project-specific operational directives; - (b) enforcing that the first component is non-overrideable by any downstream authority layer or agent configuration; - (c) enforcing that the second component is non-overrideable at the project layer; - (d) for any agent-

source artifact identified by a canonical provenance marker, routing agent context-loading unconditionally to the first component before any other context loads; and - (e) permitting downstream authority layers to define their own third-component rules only within the headroom left by the first and second components.

Claim 5 (Ring of Three Golden Eblets — Dependent: Recursive Instantiation)

The method of Claim 4, wherein each successive authority layer in a multi-layer hierarchy of at least two layers instantiates its own three-component authority structure per Claim 4, and wherein the first component of every child layer inherits and preserves the non-overrideable rules of the first component of every parent layer in the hierarchy.

Claim 6 (Multi-Layer Authority Recursion — Independent Method Claim)

A method for substrate-as-a-service deployment with compound authority-rule inheritance, comprising: - (a) defining at least two authority layers L_k and L_{k+1} in a hierarchical substrate deployment, each authority layer comprising a three-component authority structure per Claim 4; - (b) enforcing that the non-overrideable rules of L_k propagate to L_{k+1} without modification; - (c) assigning a canonical layer address to each authority layer, the address comprising the layer identifier and all parent layer identifiers in ordered sequence; - (d) validating, before any authority directive from a lower layer takes effect, that no circular authority reference exists in the layer graph that would grant a child layer authority over a parent layer; and - (e) enabling each authority layer to define its own brand identifiers, stamps, QR codes, and IP ledger entries, distinct from those of parent layers.

Claim 7 (Multi-Layer Authority Recursion — Dependent: Layer Addressing)

The method of Claim 6, wherein the canonical layer address for layer L_k at depth d in the hierarchy is expressed as a period-delimited identifier comprising the layer's own identifier followed by the identifiers of all parent layers in ascending order of authority, enabling unambiguous routing of messages and artifacts to any layer in the hierarchy by address alone.

Claim 8 (Social-Authority DAG + Pheromone-Anchored Decisions — Independent Method Claim)

A method for cross-cutting social-authority governance in a hierarchical AI substrate platform, comprising: - (a) defining one or more social-authority units, each social-authority unit being associated with a set of platform participants that spans at least two branches of the platform's hierarchical authority tree; - (b) granting each social-authority unit sovereignty over both (i) the decision-mechanism by which the unit reaches decisions and (ii) the set of participants to whom unit decisions apply; - (c) recording each social-authority unit decision as an append-only structured record in a stigmergic substrate index, the record comprising the decision identifier, social-authority unit identifier, decision mechanism used, set of participants to whom the decision applies, and a reference to one or more canonical substrate tablets without mutating those tablets; - (d) indexing recorded decisions in the stigmergic substrate index for sub-millisecond retrieval by both tablet-to-decisions and decision-to-tablets query paths.

Claim 9 (Social-Authority DAG — Dependent: Decision Supersede)

The method of Claim 8, further comprising: for each decision record, accepting a supersede field that references a prior decision record; filtering queries so that superseded decision records are excluded from active retrieval while remaining in the append-only

index for audit purposes.

Claim 10 (AGPL Federation Library — Independent Method Claim)

A method for cooperative AI substrate distribution with dual-license cost-sharing, comprising: - (a) distributing a first substrate layer under an open-source license permitting unrestricted use, modification, and redistribution without payment or membership; - (b) offering a second substrate layer providing cross-member canonical-content reuse under an optional cooperative membership, wherein the membership fee is identical for all members; - (c) computing a cost-savings multiplier for each cooperative member as a function of the number of cooperative members N and a per-member reuse rate, wherein the multiplier increases monotonically with N up to a ceiling bounded by the maximum achievable reuse rate; and - (d) applying the cooperative-member multiplier as one layer in a compound N -layer cost-savings algorithm, wherein each of the N layers provides an independent multiplicative savings contribution.

Claim 11 (Federation Library — Dependent: Five-Layer Compound)

The method of Claim 10, wherein the compound cost-savings algorithm comprises exactly five layers: (L1) vendor-routing cross-vendor uniformity, (L2) AI model-tier routing, (L3) reproducibility-based reuse, (L4) agent-spawn-boundary pre-injection, and (L5) cross-member cooperative-content reuse, and wherein the five-layer compound multiplier is the product of the five individual layer multipliers.

Claim 12 (LB Frame Broadcast Funnel — Independent Method Claim)

A method for cooperative AI substrate adoption via a staged conversion funnel, comprising: - (a) presenting a first stage offering free, one-click installation of the substrate with no payment, signup, or account creation required; - (b) automatically firing a self-verification primitive at installation completion that verifies substrate components and emits a tabular empirical receipt; - (c) presenting a second stage enabling the visitor to apply the substrate to the visitor's own data and receive a measurable output; - (d) presenting a third stage offering cooperative membership at an identical membership fee for all members, unlocking cross-member content sharing; - (e) presenting a fourth stage enabling a cooperative member to refer new visitors to the first stage under a member-issued sponsorship mark.

Claim 13 (Walkaround Preflight — Independent Method Claim)

A method for AI substrate self-verification at install time, comprising: - (a) at installation completion of an AI substrate package, automatically invoking a substrate self-verification sequence without requiring user initiation; - (b) the self-verification sequence checking each substrate component against a canonical integrity criterion and emitting, for each check, a verdict selected from {PASS, WARN, FAIL}; - (c) aggregating the per-component verdicts into a tabular receipt comprising component identifiers, verdicts, and diagnostic detail; - (d) presenting the tabular receipt to the installer as the installation completion demonstration, whereby no marketing-copy assertion about the substrate's capabilities is required; - (e) retaining the tabular receipt as an append-only Stone Tablet record for provenance.

Claim 14 (Walkaround — Dependent: Variant-Parameterized Universalization)

The method of Claim 13, wherein the substrate self-verification sequence is parameterized by a variant identifier, and wherein each variant identifier specifies a distinct component checklist appropriate to that installation variant, enabling the same

self-verification method to cover a plurality of distinct substrate installation types without modification to the verification logic.

Claim 15 (Walkaround — Dependent: Multi-Layer Integrity)

The method of Claim 13, further comprising: for substrate installations at authority layer L_k in a multi-layer authority hierarchy, the self-verification sequence additionally verifying that the cryptographic anchor hash of the installed substrate's canonical Ring of Three matches the parent layer L_{k-1} 's recorded hash for L_k , confirming that the L_k installation is integrity-anchored to its parent authority.

Claim 16 (Augur Living Gate — Independent Method Claim)

A method for substrate-event-driven freshness gating in an AI memory substrate, comprising: - (a) recording, in a stigmergic substrate index, the timestamp of each write operation to any canonical substrate record associated with a topic; - (b) recording, for each agent consultation of a topic, the timestamp of that consultation; - (c) when an agent requests a topic's canonical content, retrieving the topic's most recent write timestamp from the substrate index; - (d) if the most recent write timestamp postdates the most recent consultation timestamp, re-deriving the topic's canonical content before returning it; and - (e) if the most recent write timestamp predates the most recent consultation timestamp, returning the prior derivation without re-derivation; - (f) whereby re-derivation is triggered by actual substrate state change, not by elapsed clock time.

Claim 17 (Augur Living Gate — Dependent: Hard Ceiling Fallback)

The method of Claim 16, wherein if the stigmergic substrate index does not contain a write timestamp for the requested topic, re-derivation is triggered if and only if the elapsed time since the most recent consultation exceeds a configurable hard-ceiling duration, providing graceful degradation when the stigmergic index is unavailable.

Claim 18 (Catechist Scribe — Independent Method Claim)

A method for AI agent session-discipline auto-grading in a persistent-substrate multi-agent platform, comprising: - (a) at each AI agent session start, invoking a discipline-grading scribe via a session-start hook mechanism; - (b) the discipline-grading scribe reading the agent's first M tool calls, where M is a configurable parameter; - (c) evaluating each tool call against a predefined rule set and assigning a verdict of PASS, WARN, or FAIL with associated evidence; - (d) computing a cryptographic signature over each grade record using a keyed hash function, providing tamper-evident provenance; and - (e) writing each grade record as an append-only substrate record for retroactive audit.

Claim 19 (Persistent Background Watcher Daemon — Independent Method Claim)

A method for persistent background monitoring in a cooperative AI substrate platform, comprising: - (a) at the start of a persistent-substrate AI agent session, spawning an OS-level background process that monitors substrate state independently of the agent's main execution thread; - (b) the background process capturing substrate state changes, cross-monitor context, and session boundary events at a configurable monitoring interval; - (c) recording a process identifier for the spawned background process in a canonical per-session state file; - (d) at the termination of the AI agent session, invoking a session-end hook that reads the per-session state file and terminates the background process.

Claim 20 (PostToolUse Audit + Stone Tablet Supersede — Independent Method Claim)

A method for non-blocking AI agent discipline enforcement via post-execution audit, comprising: - (a) permitting an AI agent's tool call to execute without blocking; - (b) at tool call completion, invoking a post-execution audit hook that reads the tool call's output and evaluates it against a discipline rule set; - (c) if the audit identifies a discipline violation, writing a violation record as an append-only Stone Tablet record, and writing a supersede-pointer record at a canonical path associated with the original tool output; - (d) if the audit identifies no violation, writing a PASS record as an append-only Stone Tablet record; and - (e) providing the supersede-pointer record to a persistent-substrate agent for reconciliation at the next session boundary, without interrupting the current agent session.

Claim 21 (Substrate-Dependency Classifier — Independent Method Claim)

A method for transparent classification of commercial products by cooperative-substrate dependency level, comprising: - (a) maintaining a canonical dependency graph comprising identifiers for all cooperative substrate innovations; - (b) analyzing a commercial product's architecture, API call trace, and documentation references against the canonical dependency graph using tokenization and alias-matching; - (c) computing a dependency score as the fraction of the commercial product's commercial value estimated to derive from cooperative substrate innovations; - (d) assigning a tier classification based on the dependency score against predefined thresholds; - (e) publishing the classification algorithm, dependency graph, and output reasoning such that the classification is auditable without black-box adjudication.

Claim 22 (Detective→Wrasse Live-Update — Independent Method Claim)

A method for substrate-aware canonical retrieval with registry self-update, comprising: - (a) receiving a canonical information query at a detective scribe subsystem; - (b) the detective scribe executing the query against a stigmergic substrate index and returning a result set comprising retrieved records with confidence scores; - (c) for each result in the result set with a confidence score exceeding a threshold, automatically appending a new entry to an agent-spawn-boundary pre-injection registry associating the query's trigger pattern with the retrieved result as canonical resolution; - (d) whereby subsequent agents encountering the same trigger pattern receive the pre-injected resolution without invoking the detective scribe subsystem.

Claim 23 (Cross-Agent Edict Propagation — Independent Method Claim)

A method for cross-agent edict propagation in a heterogeneous multi-agent AI substrate platform, comprising: - (a) encoding a platform edict as a trigger-class entry in an agent-spawn-boundary pre-injection registry, associating the edict's trigger pattern with the edict's canonical resolution; - (b) at each agent session start across heterogeneous agent classes, injecting the edict's canonical resolution into the agent's context before the agent's first reasoning token fires, regardless of agent class; - (c) whereby agents across heterogeneous classes receive the same edict resolution through the same substrate mechanism, without per-class edict encoding.

Claim 24 (Compound Cooperative Architecture — Independent System Claim)

A system for cooperative AI substrate deployment with compound cost-savings and hierarchical governance, comprising: - a substrate-routed memory expansion module per Claim 3; - a hierarchical authority management module implementing the Ring of Three Golden Eblets per Claim 4 at each of at least two authority layers; - a social-authority decision-recording module per Claim 8; - a cooperative federation library module per Claim 10; - a substrate self-verification module per Claim 13; - a stigmergic substrate index (Pheromone Substrate) providing sub-millisecond retrieval for all modules above,

wherein each module writes its records to the stigmergic index and reads prior records via the same index; - wherein the compound cost-savings multiplier produced by operating all modules simultaneously exceeds the product of any subset of the individual module multipliers.

Claim 25 (Cooperative Defensive Patent Pledge — System Claim)

A system per Claim 24, wherein the system is administered under a cooperative-defensive- patent-pledge framework comprising: - a pledge that grants free use rights to all nonprofit, cooperative, and academic institutions on presentation of a verifiable registration identifier; - a pledge that restricts the applicant from using any claim in the application offensively against any party building on the cooperative substrate described in Claims 1–24; - a structural bylaw encoding the pledge as a non-adjustable governance constraint, non-overrideable by any downstream authority layer per Claims 4–7.

Claim 26 (Walkaround at Federation Member Install — Dependent Compound Claim)

The method of Claims 12 and 13, wherein the automatic self-verification at the second stage of the staged conversion funnel additionally verifies cross-member cooperative-content connectivity by confirming that the installing member's substrate can reach and authenticate with at least one remote cooperative-member substrate, and wherein the extended verification result is appended to the tabular receipt of Claim 13.

Claim 27 (Augur Gate + Living Gate Composition — Compound Method Claim)

The method of Claim 16, wherein the stigmergic substrate index is the same index used for decision recording per Claim 8, whereby social-authority unit decisions that modify a topic's canonical content automatically update the topic's write timestamp in the shared index, causing the freshness gate of Claim 16 to trigger re-derivation for all agents that consult the topic after the decision record is written.

Claim 28 (Full-Stack Compound — Independent Claim)

A method for cooperative AI substrate platform operation combining substrate-routed memory expansion, recursive authority governance, social-authority decision recording, cooperative federation cost-sharing, staged adoption funnel, and install-time self-verification, comprising executing the methods of Claims 1, 4, 6, 8, 10, 12, and 13 on a shared stigmergic substrate index such that records written by any one method are retrievable by all other methods at sub-millisecond latency, and the compound cost-savings multiplier across all methods exceeds 97% vendor-API-spend reduction at two or more cooperative members.

Abstract

A cooperative AI substrate platform implements eight architectural primitives — Substrate-Routed Memory Expansion, Ring of Three Golden Eblets Authority Structure, Multi-Layer Authority Recursion, Social-Authority DAG Pheromone-Anchored Decision Governance, AGPL Cathedral Free / Federation Library Cooperative Architecture, LB Frame 5-Stage Broadcast Funnel, Knight-Has-No- Shadows Persistent-Substrate Agent Class Distinction, and Walkaround Universal Install-Time Preflight — composing as the Cathedral Adoption Pathway. Memory expansion scales $O(\text{relevant- topics-per-task})$ rather than $O(\text{total-topics})$, demonstrated by 65% index compaction with simultaneous

60+ corpus growth (KN042). A 5-layer compound cost-savings algorithm delivers 35.72× measured savings (97.2% reduction) at N=2 cooperative members (KN056+KN057), with theoretical ceiling approaching 98.3% at scale. All innovations are filed under the Cooperative Defensive Patent Pledge (#2260): free for nonprofit, cooperative, and academic use; pledged non-offensively to all cooperative-substrate builders. Provisional filed May 1, 2026. Conversion deadline May 1, 2027.

Filing Gate Status

PUBLICATION GATE: HARD CLOSED

Per Fire Control Directive (#2330, Prov 15): this specification is internal-only pending Founder review and explicit USPTO Patent Center submission authorization.

Knight scope (KN071): build and stage this specification. COMPLETE. Founder scope: review this document; upload to USPTO Patent Center; authorize submission.

Estimated filing fee: **\$65 (Small Entity, provisional)** USPTO Patent Center:
<https://patentcenter.uspto.gov>

Mechanical gate: USPTO_SUBMISSION_AUTHORIZED=false (this specification)

KN071 call sign: v-k-prov-16-spec-consolidation-KN071 (to be applied at Phase E commit)

End of

*PROV_16_SUBSTRATE_ROUTED_MEMORY_EXPANSION_FEDERATION_DRAFT.m
d Filed: KN071 K-Prov-16-Spec-Consolidation / BP006 / 2026-05-01 by Knight Awaiting
Founder review per Fire Control Directive (#2330)*